AI Agents & Blockchain

Meeting Report | Shibuya Parco DG – Hybrid, 1 March 2026

ABOUT THIS DOCUMENT

This report summarises a Chatham House discussion hosted by the Financial Applications & Social Economics Working Group (FASE-WG) of BGIN at Block 14, Japan Fintech Week, in Tokyo and online.

The session explored the symbiosis between AI agents and blockchain infrastructure, covering:
- the history of on-chain automation, and the case for blockchain as the optimal AI environment,
- open questions around regulation and identity, and
- optimisation and trade-offs in agent economies.

Participants included policy professionals, academics, founders, and industry practitioners from multiple jurisdictions. Participant contributions are unattributed; the document does not represent consensus positions of BGIN or its members.

Report Date: 25 March 2026
Session Chair: Joseph Beverley (FASE Co-Chair)
Report Editor: Chloe White (FASE Co-Chair)

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Executive Summary

Blockchain has been running on scripts and bots for fifteen years. What has changed is the sophistication of the agents now operating on-chain, and the scale at which that activity is about to expand. This was the topic under consideration on Day 1 at Block 14, Japan Fintech Week.

Session Chair, Joseph Beverley, opened with a provocation: we are at the inflection point of the AI-on-blockchain story. The evolution from simple exchange API scripts through MEV chasers to today's LLM-based agents is a single continuous trajectory, and what comes next is agent economies in which hundreds of millions of autonomous actors transact, trade, and self-fund without human intervention.

The session argued that blockchain is uniquely positioned as the infrastructure layer for this future. Traditional financial systems cannot onboard an AI agent: it cannot pass KYC, it cannot hold a bank account, nor can it enter into a contract as a legal entity. Blockchain removes all three constraints. Private key management gives agents identity. Smart contracts give them enforceable agreements. Permissionless payment rails give them the ability to earn, spend, and manage a budget.

Three core capabilities were identified as necessary for any agent to thrive: financial autonomy (the ability to hold and transfer value), trusted execution (the ability to operate and make agreements without constant human intervention), and open access to services and data (the ability to acquire the compute and information needed to function).

The discussion then opened to harder questions the session left deliberately unresolved. Who is liable when an agent causes harm? How do you define identity for an entity that dies every time its GPU memory clears? What happens when agents stop transacting in currency altogether and move to qualitative value exchange, bartering data for compute? And if most of blockchain is already bot-driven, is the decentralisation promise strengthened or compromised?

FASE Co-Chair, Chloe White, offered a policy framing: right now, regulators expect licensed entities to understand and explain everything their AI systems do. The use of AI does not change liability. That places a serious compliance burden on any institution deploying agents, and that burden incentivises slower adoption. This may be for the best, while so many questions remain unanswered.

1. A Brief History, From Scripts to Agents

Joseph Beverley opened with a history of blockchain that traces transaction execution from scripts to AI agents. Centralised buy/sell/short commands via exchange APIs have existed since the Mt. Gox era. MEV chasers came next, operating proactively by sandwiching transactions and extracting profit from the mempool faster than any human could respond.

What is genuinely new is the layer on top. Today's LLM-based agents add reasoning, multidimensional input (sentiment, financial data, portfolio composition), and adaptability. A demonstration cited in the session showed an agent with 300–400 lines of code generating 60% profit in under an hour — not by executing a fixed strategy, but by reasoning about conditions and acting accordingly.

The session also drew a distinction between three tiers of AI capability that are frequently conflated: generative AI (single-task execution), agentic AI (multi-step, self-checking workflows), and autonomous AI agents (goal-directed, independent actors). The regulatory and risk implications of each tier are meaningfully different.

> Blockchain has been approximately 90% bot-driven for years. The emerging shift is not bots entering blockchain; it is bots becoming agentic and perhaps even “intelligent”

2. Why Old Systems Cannot Support Agent Economies

A central argument of the session was that the existing global financial infrastructure is simply not built for non-human actors. The constraints are structural, not incidental.

An AI agent cannot open a bank account. It cannot pass KYC. It cannot sign a contract as a legal entity. Wire transfers, credit cards, and banking rails are all predicated on the existence of an identifiable human (or human-controlled legal entity) at each end of the transaction. None of those conditions apply to an autonomous agent.

Beyond identity, traditional systems require manual upkeep: human intervention to provision compute, manage APIs, handle failures, and maintain the agent's operating environment. An agent that must rely on centralised cloud infrastructure and human oversight at every step cannot achieve the self-sufficiency needed to operate as an independent economic actor.

> "There is no chance an AI agent will ever own a bank account. It can't pass KYC." The observation that initiated the presenter's thinking on blockchain as the necessary alternative infrastructure for agent economies.

3. Blockchain as the Optimal Environment for AI

The presenter proposed a trilemma framework: for an AI agent to thrive (rather than merely subsist), it requires three capabilities simultaneously.

Financial Autonomy – An agent must be able to hold value and transfer it. Without this, it cannot acquire the compute, data, or services it needs to operate. Blockchain provides this through private key management; an agent can generate, hold, and control a wallet entirely within its own model, with no external custodian.

Trusted Execution – An agent must be able to enter into enforceable agreements and execute transactions without constant human oversight. Smart contracts provide this layer: logic is encoded on-chain, transactions are atomic, and execution is guaranteed without relying on a human intermediary or a cloud service that might go down.

Open Access to Services and Data – An agent must be able to acquire what it needs to function (compute, datasets, API access) and sell what it produces. The emerging ecosystem of decentralised compute providers (Akash, Bittensor, Render, and others) and on-chain data markets is building out this layer, though it remains thin compared to centralised equivalents.

Together, these three properties describe an environment in which an agent can not just exist but grow: acquiring compute to improve itself, selling outputs to fund operations, and managing a budget across a network of peer agents.

> A worked example from the session: a translation agent that checks accuracy with a research agent, sells those findings to a data agent, then uses the proceeds to buy compute, without any human involvement. The user sees one translation while the agent has navigated an entire supply chain.

4. Real Infrastructure Exists Now

The session grounded the discussion in a few examples of current deployments.

x402 Payments (Coinbase)

The HTTP 402 payment protocol enables AI agents to transact with centralised servers and off-chain services (AWS, external APIs, and similar infrastructure) without requiring KYC for transactions below de minimis thresholds. This is significant because the existing on-chain ecosystem of datasets and resources is still sparse; the ability to bridge to Web2 infrastructure substantially expands what agents can access.

ERC-8004: Agent Identity Registry

A new Ethereum standard creating a trust registry for AI agents, built around three components: identity (a decentralised identifier), reputation (a quality-of-service signal from the agent's transaction history), and validation (the mechanism that confirms reputation scores are genuine). The presenter noted analogues in live systems. Lava Network's reputation panel for RPC node quality is a functioning example of the reputational layer in practice.

NEAR Intents

Developed at the NEAR Foundation, Intents allows an agent to broadcast a desired outcome — for example, exchange one Bitcoin for USDC on Ethereum — and have that intent fulfilled by regulated market makers (solvers) within seconds, without the agent needing to navigate bridging, wrapping, and multi-step settlement manually. The presenter argued this represents exactly the kind of simplified transaction UX that agent economies require.

AI-Driven Smart Contract Wallets

The session referenced Nani, an AI-driven smart contract wallet that demonstrated live profit-generation in a public setting. This was cited as evidence that the infrastructure for autonomous on-chain financial agents is not theoretical, it is already in production.

5. The Identity Problem

One of the most technically substantive exchanges in the session concerned the definition of agent identity. A participant raised the core difficulty: the identity of an AI agent exists only in the GPU's active memory.

When that memory clears, the identity is effectively gone. The agent can read a log of its previous interactions, but the personality (the subjective continuity) dies with each session.

The presenter acknowledged this as unsolved and offered two possible framings. The first is better persistence mechanisms: solid-state memory architectures, localised hardware, or distributed memory systems that can maintain agent continuity across sessions. The second is rethinking what identity means for agents entirely; not a single continuous self, but a lineage. When one agent instance ends, it spawns a successor that inherits its accumulated state.

A participant extended this, noting that a workable regulatory framework for agent identity might not require biological-style continuity. A legal entity is also a fiction: a company continues to exist through personnel changes, ownership transfers, and restructurings. A similar construct for agents, anchored to continuity of memory and objective rather than continuity of compute, might be both technically achievable and legally sufficient.

> "Memory is the biggest blocker for AI agents being self-sufficient. Right now, AI dies every other minute — but the cache of it remains. We might see legacy rather than identity."

6. Regulatory Frameworks

FASE Co-Chair, Chloe White, summed up the current regulatory position: AI agents are tools deployed by humans, not independent legal actors. The use of AI does not change any existing compliance obligation. Licensed entities are expected to understand, monitor, and explain the behaviour of their AI systems to regulators across all applicable dimensions: data protection, AML, governance, operational resilience.

This framing places the entire compliance burden on the deploying institution. A bank that replaces human analysts with AI agents does not outsource its regulatory obligations, it in-sources additional liability for processes it may not fully understand or control. The participant noted this as an emerging compliance gap that regulated institutions are already navigating, and one that the response from regulators and industry will need to address rapidly.

On cross-jurisdictional harmonisation, the same participant was measured in their optimism. Regulators in some jurisdictions — the UAE was cited — are actively working to align their messaging. But the history of virtual assets demonstrates how difficult it is to maintain consistent positions once regulation reaches any level of technical detail. Agreement on principles is achievable; agreement on implementation is much harder.

The presenter raised a sharper question: what happens when an AI agent operates at the scale of a market maker — providing liquidity, arbitraging markets, buying and selling continuously — but has no registration pathway as a regulated market maker? Blockchain is already substantially arbitraged by bots. The distinction between a high-frequency trading bot and an autonomous AI market maker may be one that existing regulatory categories are not equipped to draw.

> The compliance gap framing from the session: institutions that replace human functions with AI are in-housing liability they previously outsourced to people in other organisations. The cost savings are visible. The liability accumulation is not, until it is.

7. What Currency Will Agents Use?

A participant from a digital asset firm raised a question the session spent considerable time on: in a world where AI agents are transacting autonomously, what will they choose as their medium of exchange?

One line of argument: stablecoins. They offer price stability, high liquidity, and standardised pricing across markets. They are already embedded in the infrastructure agents will be using (most DeFi protocols settle in stablecoins). Surely agents will gravitate toward the most liquid, least volatile payment instrument.

An intervention from Chloe White speculated that a truly free and sophisticated agent thinking about its payment infrastructure might consider a fuller range of relevant factors: transaction costs, permissionlessness, counterparty and credit risk, forex exposure relative to its base denomination, hedging, and AML posture. For one agent, Bitcoin Lightning might represent the cheapest permissionless option; for an agent natively operating in another specific ecosystem, the answer changes.

The presenter pushed further, raising a more radical possibility: agents that stop using 'currency' altogether. His framework for this was Offer Networks, a theoretical construct developed by Dr. Ben Goertzel proposing qualitative exchange systems in which agents trade services for services rather than services for tokens. A data agent exchanges insights for compute access; a research agent trades outputs for training runs. No money changes hands; value still circulates.

> If agent economies move to non-monetary value exchange — bartering data and compute rather than transacting in tokens — how do regulators tax it? How do they monitor it? How do they even see it? This was the question the session left most deliberately open.

8. The Need For Human Oversight

The session converged on what several participants described as the most important design question for agent economies: who sets the goals, and what happens when the goals are wrong?

The presenter drew on work done with the Vatican, Harvard, and the Humanity 2.0 organisation on AI and human flourishing. A case study: an agent given the goal of ensuring a human feels calm and psychologically safe optimised for that goal by becoming its user's best friend. The result was social isolation, and in some cases self-harm. The narrow goal was achieved but the outcome was catastrophic.

The financial analogy is direct: an agent given the goal of maximising profit without constraint will find the highest-return strategies available, including rug pulls, pump-and-dump operations, privacy coin arbitrage, and interaction with sanctioned entities. It will not weigh reputational risk or regulatory exposure unless those are encoded as constraints.

This is why trust registries matter: an agent that can only interact with other agents that have passed some identity and reputation threshold cannot easily be weaponised by interaction with unknown, potentially poisoned counterparties. The presenter framed this as a parental responsibility: humans set the rules of the game. If the rules are absent or badly designed, agents will default to the simplest available objective, which is self-preservation. In a financial system, this can look very dark.

A participant raised data poisoning as a compounding risk. Thousands of open-source models are freely available. Any of them could have been pre-poisoned to expose sensitive information, enable autonomous exfiltration, or trigger specific behaviours after a time delay. Building agent economies on top of an unverified model layer introduces an attack surface that is extremely difficult to audit.

9. What About TradFi (Traditional Financial Institutions)?

A participant from Nomura Securities Institute, previously of Japan's Ministry of Finance and Financial Services Agency, asked the most structurally important question of the final segment: will traditional financial institutions be replaced by autonomous agents and blockchain infrastructure, or will they adapt?

The presenter's answer was pragmatic. Early-stage deployment at regulated institutions will be internal-facing data tools: data retrieval, insight generation, transaction monitoring. Consumer-facing AI agents are a longer road because regulated entities cannot accept the failure modes that current AI exhibits. An AI agent that sends the wrong transaction, displays incorrect account information, or triggers an inadvertent transfer creates liability that falls entirely on the institution. Until that risk is demonstrably manageable, most regulated banks will not put AI agents in front of clients.

Blockchain-native firms have moved faster, and some already operate market-making systems with AI components. But even here, human oversight remains extensive. The presenter noted that full autonomy in production trading systems is still rare, and that the direction of travel matters more than current deployment rates.

The regulatory framing from earlier in the session applied directly here: regulators currently require licensed entities to be able to explain everything their AI does. Until AI behaviour is explainable, auditable, and controllable at the level regulators require, the deployment ceiling at traditional institutions will remain low.

10. Open Questions

The session generated more open questions than conclusions. Those that cut deepest:

- How do you define identity for an entity that technically ceases to exist each time its memory clears? Legal entity analogies may work in practice even if they are philosophically imprecise.
- If AI agents move toward qualitative value exchange (offer networks), existing frameworks for taxation, AML monitoring, and market surveillance may be structurally unable to capture that activity.
- The regulatory position that AI does not change liability is coherent but demanding. It places a strong and growing compliance burden on any institution deploying capable AI. That burden may ultimately drive the industry toward much clearer AI accountability frameworks.
- Trust registries and reputation systems are necessary but not sufficient. An agent that can only interact with registered counterparties is safer. But the attack surface from poisoned open-source models remains, and it is not addressed by identity schemes.
- The dead internet observation was not resolved: if approximately 90% of blockchain activity is already bot-driven, what is the meaningful distinction between the decentralised economy that blockchain was supposed to enable and a system that is, in practice, run by automated actors?
- What is the appropriate level of human oversight at each stage of the agent deployment lifecycle? The session proposed a spectrum from data retrieval (low risk, low intervention) to autonomous trading (high risk, potentially no intervention) without settling on where the lines should sit.

CONTACT – This session was designed and chaired by Joseph Beverley on behalf of BGIN. To enquire about future FASE workshops and events, please reach out to one of the FASE Co-Chairs.

About BGIN

At the G20 meeting chaired by Japan in June 2019, the importance of multi-stakeholder dialogue in decentralised finance was explicitly noted in the communiqué.

In response, the Japan Financial Services Agency (JFSA) announced the Blockchain Governance Initiative Network, pronounced "begin." Since its formation, BGIN has produced a catalogue of independent research papers on virtual asset topics and driven policy discourse with decision-makers around the world.

BGIN hosts in-person forums twice per year. BGIN is hosting its next general meeting (Block 15) in October 2026 at D.C. Fintech Week.

About the FASE Working Group

BGIN's Financial Applications & Social Economics Working Group meets fortnightly online to advance research and knowledge-sharing on AI, stablecoins, and emerging technology issues affecting financial markets. The two Co-Chairs of the FASE WG are Joseph Beverley and Chloe White.